5.12. MIXED DECAY

A radioactive sample may contain several different radioactive nuclides which are not genetically related. The decay of each nuclide follows the decay equations of the previous section. The detector measures a certain amount of the radioactivity of each species so that

$$R = R' + R'' \tag{5.66}$$

which with the introduction of (5.59) and (5.64) gives

$$R = \psi' A'_0 \, e^{-\lambda' t} + \psi'' A''_0 \, e^{-\lambda'' t} \tag{5.67}$$

Figure 5.10 shows the composite decay curve for the mixture of ^{71}Zn ($t_{½}$ 3.9 h) and ^{187}W ($t_{½}$ 23.8 h). If the half-lives of the species in the mixture differ sufficiently, as in this

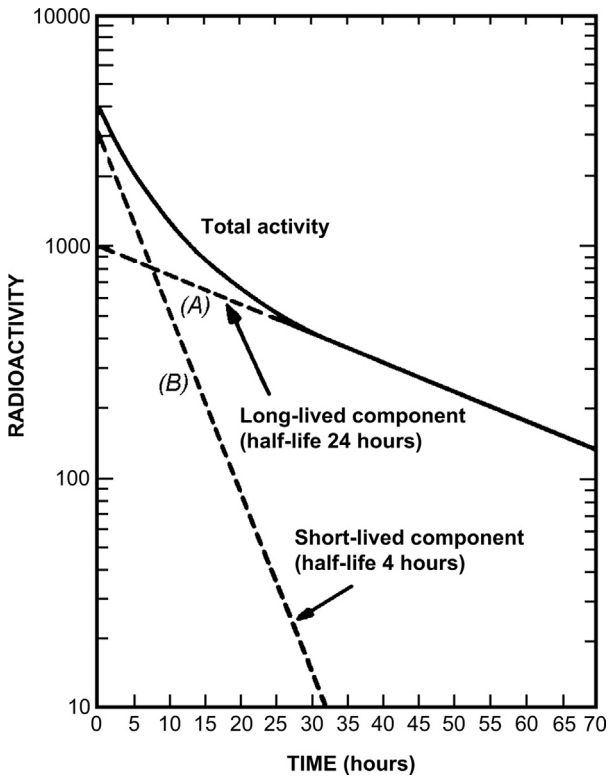


Figure 5.10 Decay diagram of a mixture of two independently decaying nuclides with half-lives of 4 and 24 h.

case, the decay curve can be resolved into the individual components. The long-lived nuclide (^{187}W, line A) can be observed to have linear decay at times long enough for the shorter-lived species to have died. The decay line related to this species can be extrapolated back to $t = 0$ and this line subtracted from the observed decay curve. The resulting curve should be the linear decay due to the shorter-lived species in the sample (line B). For more complex mixtures, this process may be repeated until the curve is completely resolved into linear components.

Sometimes it is possible in mixed decay to observe preferentially the decay of one species by proper choice of detection technique. For example, a proportional counter may be used at an operating voltage that allows detection of α-decay ($\psi_\alpha > 0$) but excludes detection of β-decay ($\psi_\beta = 0$). By contrast a typical Geiger counter can be used for β-decay but does not detect α-radiation since the α-particles do not penetrate the window of the Geiger tube. These problems are discussed more extensively in Ch. 9.

5.13. RADIOACTIVE DECAY UNITS

The SI unit for radioactivity is the *Becquerel (Bq)*, and the activity is given in reciprocal seconds, s^{-1}:

$$1 \text{ Becquerel } (\text{Bq}) = 1 \text{ } (\text{disintegration}) \text{ } s^{-1} \tag{5.68}$$

The *measured count rate* (R, in (5.64)) is given in *counts per second* (cps) or per minute (cpm); the abbreviations are in German Impulse pro minute (Ipm), and in French coups de minute (c/m).

An earlier unit, still in some use, is the *Curie unit* (abbreviated Ci) defined as:

$$1 \text{ Curie } (\text{Ci}) = 3.7 \times 10^{10} s^{-1} (\text{Bq}) \tag{5.69}$$

The Curie unit was originally defined as the number of decays per unit time and gram of ^{226}Ra, assuming its half-life to be 1580 y.

The *specific radioactivity* S is defined as the decay rate A per unit amount w of an element or compound,

$$S = A/w \tag{5.70}$$

The SI unit of specific radioactivity is Bq kg^{-1}. For practical purposes it is sometimes also defined in dpm g^{-1} or dpm $mole^{-1}$. *Activity concentration* (or "radioactive concentration") is given in Bq m^{-3} or Bq l^{-1}. With the half-life of 1599 ± 4 y the specific activity per gram of ^{226}Ra is 0.988 Ci or 3.65×10^{10} Bq or 2.19×10^{12} dpm. The specific activities of some of the longer-lived naturally occurring radioactive species are: ^{40}K, 31.0 kBq kg^{-1} of K; ^{232}Th, 4.06 MBq kg^{-1} of Th; ^{238}U, 12.3 MBq kg^{-1} of U.